Xuhan Tong

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EDUCATION

Tsinghua University, Beijing, China 08/2023 – 07/2027 (Expected)
Bachelor of Science in Mathematics and Physics + Software Engineering (Double Major)
GPA: 3.9/4.0
Rank: 3/37

PUBLICATIONS

Tong, Zeng, Zhang (2026). Demonstrations, CoT, and Prompting: A Theoretical Analysis of In-Context Learning. The 2026 Conference on Language Modeling Workshop on Scientific Understanding of Foundation Models. https://arxiv.org/abs/2603.19611

Hu*, **Tong***, Bai*, et al. (2026). Understanding Why Language Models Hallucinate: Testing Reasoning Against Priors. https://arxiv.org/abs/2607.00447

RESEARCH EXPERIENCE

Multi-Agent Collaboration under Multiple Constraints 07/2026 – present
Research Assistant, Prof. Jiawei Zhang, University of Wisconsin-Madison

- Develop a multi-agent collaboration framework in which specialized agents coordinate task allocation, information sharing, and decision-making under heterogeneous and potentially conflicting constraints.
- Investigate how communication protocols and coordination strategies affect constraint satisfaction, solution quality, and robustness in complex multi-step tasks.

Detecting and Mitigating Hallucination via Counterfactual Cue Interventions 05/2026 – present
Research Assistant, Prof. Jiawei Zhang, University of Wisconsin-Madison

- Develop a formal detection theory on the latent key model with a contrast identity linking observable log-odds shifts.
- Design an automated pipeline for cue extraction, validated counterfactual perturbation, and detection, which beats uncertainty baselines and separates knowledge-deployment failures from ignorance.
- Extend the framework into a stressor-symptom diagnostic model spanning four failure sources; show via mutual-information analysis that surface error symptoms are only weakly diagnostic while hidden-state probes recover the underlying cause, and that mitigation efficacy is highly cause-specific.

Understanding Why Language Models Hallucinate: Testing Reasoning Against Priors 02/2026 – 05/2026
Research Assistant, Prof. Jiawei Zhang, University of Wisconsin-Madison

- Developed a theoretical framework that explains hallucinations across different prompting and inference settings by deriving theoretical bounds connecting hallucination to generalization error.
- Modeled LLM inference using a latent key–task structure consisting of keys (salient prompt patterns) and tasks (underlying reasoning objectives).

- Identified frequency as the source of hallucination by showing that high-frequency key–task pairs dominate posterior inference, causing errors when the model selects a statistically dominant but incorrect task.
- Designed a benchmark to study hallucination mechanisms and demonstrated a strong correlation between hallucination errors and frequency imbalance in key–task distributions.

Demonstrations, CoT, and Prompting: A Theoretical Analysis of In-Context Learning 10/2024 – 03/2026
Research Assistant, Prof. Jiawei Zhang, University of Wisconsin-Madison

- Constructed a Lipschitz-based theoretical framework for ICL generalization and theoretically proved a generalization bound w.r.t. the Lipschitz constant.
- Provided a principled explanation of demonstration effectiveness by analyzing the minimum Lipschitz constant along paths connecting test prompts to pretraining samples.
- Modeled Chain-of-Thought (CoT) prompting as task decomposition into multiple subtasks and proved that CoT improves ICL when the Lipschitz constants correspond to subtasks satisfying certain conditions.
- Proved that prompt template influence decays exponentially as the number of demonstrations increases.

Prediction and Parameter Optimization of Myopia Control Effect in Smart Orthokeratology 10/2024 – 04/2025
Research Assistant, Prof. Wenkai Lu, Tsinghua University

- Built a clinical patient database and implemented a Standardized Euclidean Distance algorithm for similar-case matching and automated lens parameter recommendations.
- Developed corneal topography prediction and segmentation models to support long-term assessment and parameter optimization for myopia control.

HONORS & AWARDS

- Comprehensive Outstanding Scholarship, Tsinghua University, 2024-2025
- Science and Technology Innovation Outstanding Scholarship, Tsinghua University, 2023-2024
- Academic Excellence Scholarship, Tsinghua University, 2023-2024
- Third Prize, 43rd “Challenge Cup” Information Technology Track, Tsinghua University, 04/2025
- Second Prize, 1st “Zhili Cup” Agent Competition, Tsinghua University, 09/2024

SERVICE

- Vice President, Zijing College Association of Science and Technology, Tsinghua University
- Vice Chair, Department of Culture, Weiyang College Student Union, Tsinghua University

SKILLS

- Programming Languages: Python, C, C++, SQL, JavaScript, HTML
- Standardized Tests: TOEFL 115 (R28, L29, S30, W28), GRE 331 (V161, Q170, AW 3.5)